Examiner only

**7.** Volac is a dairy food manufacturing company based in Felinfach in West Wales.
Volac use a combined heat and power biomass power station.
It produces some of the electricity and heat for the factory.
The biomass power station burns wood produced locally.
Data about the biomass power station is given below.

| |
|---|
| Cost to build the biomass power station = £38 million (£38 000 000)<br>Estimated annual savings on energy bills = £3.75 million (£3 750 000) |

(a) (i) Calculate the expected payback time for the biomass power station. [2]

payback time = .................................. years

(ii) State **one** reason why this payback time may change. [1]

..............................................................................................................................................

(b) Seren states that burning wood produces $CO_2$ so it is harmful to the environment.
Explain whether you agree with Seren. [2]

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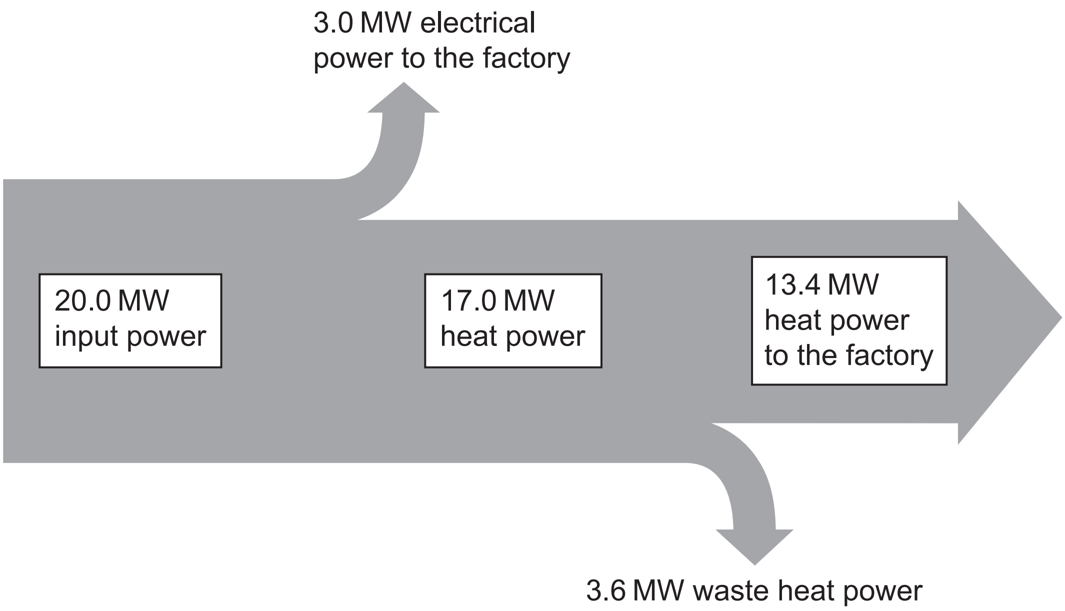
Examiner only

(c) A Sankey diagram for the biomass power station is shown below.




(i) Calculate the **total** useful output power from the biomass power station. [1]

useful output power .............................. MW

(ii) Evan states that the biomass power station is only 15% efficient.
Explain whether you agree. Include a calculation. [2]

..........................................................................................................................................................................



Examiner only

(d) (i) The input power to the biomass power station is 20.0 MW.
Use an equation from page 2 to calculate how much energy is transferred by burning wood in the power station every 60 minutes.
Give your answer in MJ. [3]

energy transferred every 60 minutes = .............................. MJ

(ii) Burning 1 tonne of the wood used in the biomass power station produces 2880 MJ of energy.
Calculate how many tonnes of wood the biomass power station burns every 60 minutes. [1]

mass of wood burned every 60 minutes = .................................... tonnes

(iii) The wood used in the biomass power station has a density of 500 kg/m$^3$.
Use the equation:

$$\text{volume} = \frac{\text{mass}}{\text{density}}$$

to determine the volume of wood burned every 60 minutes.
1 tonne = 1000 kg. [2]

volume of wood burned every 60 minutes = .................................. m$^3$

(iv) An average tree produces 5 m$^3$ of wood.
Calculate how many trees are burned every 60 minutes. [1]

number of trees burned every 60 minutes = ...........................

| |
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| |
| 15 |

**END OF PAPER**



 